Algebra 1 - Q2.1_2.3 A

Name: ____

2.1 – Expressions and Their Parts

1. I can identify the parts of an algebraic expression

In the following expression, list any **terms**, **coefficients**, and **constants**.

$$3x^2 - 2x + 5$$

Terms:

Coefficients:

Constants:

2. I can describe what a variable, constant, coefficient, and term are

Answer each question about the expression $5x - 2y + 3x^2 - 9$ with **true** or **false**.

a. The expression has 3 terms __________

b. The coefficients are 5, -2, and 3 _________

c. The constant is 9 _________

d. Both x and y are variables in the expression _________

2.2 – Evaluating Expressions

3. I can substitute values into expressions and simplify them

Evaluate the following expressions.

a. $5x - 7$ when $x = -2$

b. $-a^2 + 3b - ab$ when $a = -4$ and $b = 2$

4. I can use the correct order of operations with variables and parentheses

Evaluate the following expressions.

a. $2(x-1)^2 + 5x$ when $x = 2$

b. $3|2y - x| + 5y$ when $x = 8$ and $y = -10$

c. $\frac{1}{x}(2x-3)^2 + x$ when $x = 3$

2.3 – Combining Like Terms

5. I can identify like terms in an expression

For each of the following, indicate whether the terms are like terms (**true** or **false**).

a. $5x$ and $-5x$ ________

b. $-x^2$ and $-x^3$ ________

c. $2xy$ and $3yx$ ________

d. $5x^2y$ and $-9xy^2$ ________

6. I can combine like terms to simplify an expression

Simplify the following expressions by combining like terms.

a. $52 - 3x^2 + 5x - 11x + 3$

b. $a + 2b - 3a + b^2 - 5 + 2b^2 - 1$